

Existence of traveling waves for the nonlocal Burgers equation

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Abstract

We study the equation $u_t + uu_x + u - K * u = 0$ in the case of an arbitrary $K \geq 0$, which is a generalization of a model for radiating gas, in which $K(y) = \frac{1}{2}e^{-|y|}$. Using a monotone iteration scheme argument we establish the existence of traveling waves, which gives a solution to an open question raised by D. Serre [L^1 -stability of nonlinear waves in scalar conservation laws, in: *Evolutionary Equations*, in: *Handb. Differ. Equ.*, vol. I, North-Holland, Amsterdam, 2004, pp. 473–553].

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1. Introduction

The system of equations

$$\begin{cases} u_t + uu_x + q_x = 0, \\ -q_{xx} + q + u_x = 0, \end{cases} \quad (1.1)$$

for $x \in \mathbb{R}$ and $t \geq 0$ was derived by Hamer in 1971 [2, eqs. (4.16), (4.19)] (see also [3]). Namely, he considered the Euler system of conservation of mass, momentum and energy for a radiating gas, in which the heat flux q is given by an integral equation. He then perturbed the system about an equilibrium state and the speed of sound, and obtained (1.1) as the first-order approximation.

The system (1.1) is formally equivalent to

$$u_t + uu_x + u - \frac{1}{2} \int_{\mathbb{R}} e^{-|x-y|} u(y, t) dy = 0, \quad (1.2)$$

and can be considered a special case of the integrodifferential equation

$$u_t + uu_x + u - K * u = 0, \quad (1.3)$$

where $K * u(x, t) \equiv \int_{\mathbb{R}} K(x - y) u(y, t) dy$, K is even and $\int_{\mathbb{R}} K = 1$. Nonlinear equations with a nonlocal operator considered instead of the Laplacian have a long history and have recently been gaining increasing popularity, due

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to superior physical qualities of such models and at the same time often ingenious mathematical solutions. In fact, when in 1893 J.D. van der Waals derived his variational thermodynamic theory of binary phase transitions, he first introduced a mean field K , formally expanded $K * w - w$ into a power series: $\int_R K(y)(w(z+y) - w(z))dy = c_1 w''(z) + c_2 w^{iv} + \dots$, where $c_1 = \frac{1}{2} \int_R y^2 K(y)dy$, $c_2 = \frac{1}{24} \int_R y^4 K(y)dy$, etc., and used just the first term in this expansion [4]. He also investigated his nonlocal model with the Green's function $K(y) = \frac{1}{2}e^{-|y|}$ [4].

The Hamer equation (1.2) was studied in [2,5–8] and the more general nonlocal Burgers equation (1.3) in [9,10,1]. In particular, it was discovered that the solution of the Cauchy problem for (1.3) with smooth initial data is either smooth for all times or becomes discontinuous in finite time. Here we focus on traveling wave solutions $U(x-st)$ of (1.3). To account for possible shock discontinuities, a traveling wave is defined as a weak solution, i.e., a function U which satisfies

$$\int_R \left[\left(\frac{1}{2}U^2 - sU \right) \phi' + (K * U - U)\phi \right] = 0 \quad (1.4)$$

for an arbitrary $\phi \in C_0^\infty$. In [6] the authors obtained the following result (see also a partial version in the earlier [5]).

Proposition 1.1 ([6]). *For every pair $u_- > u_+$ there exists a solution $U(x-st)$ of (1.4) with $K(y) = \frac{1}{2}e^{-|y|}$, $U(-\infty) = u_-$, $U(+\infty) = u_+$ and $s = \frac{1}{2}(u_- + u_+)$. Moreover, (a) when $u_- - u_+ > \sqrt{2}$, U is continuous except for one point, (b) when $u_- - u_+ \leq \sqrt{2}$, U is C^1 , (c) when $u_- - u_+ \leq \frac{2\sqrt{2}k}{k+1}$, U is C^k .*

Subsequently, Denis Serre raised the question of whether a similar result can be established for (1.3) [1, Open Problem 1, p. 33]. In Section 2 we answer it in the affirmative, though we fall short of providing an exact regularity criterion of the profiles as in Proposition 1.1. Also, the method does not cover the cases when K is not even or not nonincreasing on $(0, \infty)$, or when in (1.3) one has $f'(u)u_x$ instead of uu_x , where f is a general convex function. The assumption $K(y) = o(\frac{1}{y^4})$ as $y \rightarrow \infty$ could be relaxed, if a subsolution different from the $s(x) = -\frac{2u_c}{\pi} \tan^{-1}(\epsilon x)$ below is found — the reason being that it has to have slower convergence rates to $\pm u_c$ as $x \rightarrow \mp \infty$ than s has, since the convergence rate of a solution depends on K .

2. Existence of traveling waves

Theorem 2.1. *Let $\int_R K = 1$, K even, $K \geq 0$, $K \in W^{1,1}(R)$, K nonincreasing on $(0, \infty)$, $K(y) = o(\frac{1}{y^4})$ as $y \rightarrow \infty$. For every pair $u_- > u_+$, there exists a solution $U(x-st)$ of (1.4) such that $U'(x) < 0$ and $U''(x) < 0$ for $x \neq 0$, $U(-\infty) = u_-$, $U(+\infty) = u_+$ and $s = \frac{1}{2}(u_- + u_+)$. If U is continuous, then it is also C^1 . Moreover, if $u_- - u_+ > 4 \int_R |y|K(y)dy$, then U is discontinuous at one point.*

Proof. Proposition 1.1 was proved by a phase-plane analysis [6]. Here we argue in an indirect way as we construct a monotone iteration scheme and pass to the limit. This is a standard elliptic technique, sometimes called the sub-supersolution method, whose use in nonlocal problems dates from [11] (see also the many follow-up papers on traveling waves in various nonlocal monostable models). In an early and unpublished version of [12] the authors used it to construct continuous and discontinuous stationary waves of the nonlocal bistable equation [13]. A similar argument appeared in [14] for the bounded interval case.

The main difficulty is in setting up the difference scheme in such a way that the sequence of monotone solutions is monotone. The whole sequence then has a limit so that one can pass to the limit on both sides of the scheme to obtain a solution (this procedure would not work if only a subsequence was convergent). To ensure that the solution has a preimposed limit at $-\infty$, one also needs to find an appropriate subsolution.

It suffices to show that for any constant $u_c > 0$,

$$uu' = K * u - u \quad (2.1)$$

admits a monotone and odd solution u such that $u(-\infty) = u_c$ and $u(+\infty) = -u_c$. If we let $U = u + s$, where $u_c = \frac{1}{2}(u_- - u_+)$ and $s = \frac{1}{2}(u_- + u_+)$ (the Rankine–Hugoniot condition), then, since $\int_R K = 1$, U satisfies

$$U'(U - s) = K * U - U,$$

$$U(-\infty) = u_- \text{ and } U(+\infty) = u_+.$$

We will see that it suffices to study (2.1) on $(-\infty, 0)$. We find it convenient first to find a supersolution and a subsolution for (2.1). Let

$$u_0(x) \equiv \begin{cases} u_c, & x < 0, \\ 0, & x = 0, \\ -u_c, & x > 0. \end{cases}$$

Then it is easily seen that

$$u_0 + u_0 u'_0 \geq K * u_0$$

on $(-\infty, 0)$; thus u_0 is a supersolution there.

Let

$$s(x) \equiv -\frac{2u_c}{\pi} \tan^{-1}(\epsilon x).$$

Note that

$$g(x, \epsilon) \equiv \frac{-\int_R K(x-y)[\tan^{-1}(\epsilon y) - \tan^{-1}(\epsilon x)]dy}{\frac{2u_c}{\pi} \tan^{-1}(\epsilon x) \frac{\epsilon}{1+\epsilon^2 x^2}} \leq 1 \quad (2.2)$$

for $x < 0$ is equivalent to s being a subsolution on $(-\infty, 0)$, i.e.,

$$s + ss' \leq K * s.$$

We show that (2.2) holds for small enough $\epsilon > 0$. Let $z = \epsilon x$ and consider $h(z, \epsilon) \equiv g(\frac{z}{\epsilon}, \epsilon)$ separately on three intervals: $(-\infty, z_2]$, $[z_2, z_1]$, $[z_1, 0]$. Using l'Hôpital's rule, we get

$$\lim_{z \rightarrow 0} h(z, \epsilon) = \frac{\epsilon \pi}{2u_c} \int_R \frac{y^2 K(y)}{1 + \epsilon^2 y^2} dy,$$

which is $O(\epsilon)$ since $\int_R y^2 K(y) dy < \infty$. Also,

$$\lim_{z \rightarrow -\infty} h(z, \epsilon) = \lim_{z \rightarrow -\infty} \frac{1}{u_c} \left[\frac{\epsilon}{2z} \int_R \frac{y^2 K(y)(1+z^2)}{1 + (\epsilon y + z)^2} dy + \int_R \frac{y K(y)(1+z^2)}{1 + (\epsilon y + z)^2} dy \right] = 0.$$

Observe that if any $\frac{h_1}{h_2}$ has a maximum at a point, then at that point $\frac{h_1}{h_2} = \frac{h'_1}{h'_2}$. Thus it is enough to estimate $\frac{h'_1}{h'_2}$. Doing so first for small z , we can neglect $\frac{1}{1+z^2}$ in the denominator of h :

$$\frac{h'_1}{h'_2} = \frac{\pi}{2u_c} \left[\epsilon \int_R y^2 K(y) dy + z \int_R \frac{2y K(y)}{1 + (\epsilon y + z)^2} dy \right],$$

and thus there a z_1 such that for all small $\epsilon > 0$ and $z \in [z_1, 0)$, $h \leq 1$. For large z , we can neglect $-\tan^{-1} z$ in the denominator of h :

$$\frac{h'_1}{h'_2} = \frac{\pi}{2u_c} \left[\frac{\epsilon}{2z} \int_R y^2 K(y) \frac{1+z^2}{1 + (\epsilon y + z)^2} dy + \int_R y K(y) \frac{1+z^2}{1 + (\epsilon y + z)^2} dy \right]. \quad (2.3)$$

In the second integral in (2.3) we break up R into two sets: $[\frac{-z}{2\epsilon}, \frac{-2z}{\epsilon}]$ and $R \setminus [\frac{-z}{2\epsilon}, \frac{-2z}{\epsilon}]$. We get

$$\int_{\frac{-z}{2\epsilon}}^{\frac{-2z}{\epsilon}} y K(y) \frac{1+z^2}{1 + (\epsilon y + z)^2} dy \leq \frac{3z^2}{4\epsilon^2} K\left(-\frac{2z}{\epsilon}\right) (1+z^2), \quad (2.4)$$

and thus from $K(y) = o(\frac{1}{y^4})$ as $y \rightarrow \infty$ there is a $z_2 < 0$ such that for $z \leq z_2$ the right side of (2.4) is small uniformly in ϵ . After integration by parts the second integral can be estimated from

$$\left| \int_{R \setminus [\frac{-z}{2\epsilon}, \frac{-2z}{\epsilon}]} y K(y) \frac{1+z^2}{1 + (\epsilon y + z)^2} dy \right| \leq -Q\left(\frac{-z}{2\epsilon}\right) \frac{1+z^2}{1 + \frac{z^2}{4}} - Q\left(\frac{-2z}{\epsilon}\right) + 8\epsilon \int_R |y| K(y) dy,$$

where $Q(y) = \int_{-\infty}^y rK(r)dr = o(\frac{1}{y^2})$ as $y \rightarrow \infty$; thus it is small for $z \leq z_2$ and ϵ small. The first integral in (2.3) is dealt with in a similar way, so altogether $h(z, \epsilon) \leq 1$ for $z < z_2$ and all small $\epsilon > 0$. Finally, for $z \in [z_1, z_2]$ define $h(z, \epsilon) = 0$. Using l'Hôpital's rule, we get

$$\lim_{\epsilon \rightarrow 0} h(x, \epsilon) = \lim_{\epsilon \rightarrow 0} \frac{\pi}{2u_c} \frac{1+z^2}{\tan^{-1} z} \int_R \frac{yK(y)}{1+(\epsilon y+z)^2} dy = 0,$$

and thus using uniform continuity, for $\epsilon > 0$ small enough, we also have $h(z, \epsilon) \leq 1$.

On $(-\infty, 0)$ define the sequence $\{u_n\}_{n \geq 0}$ by

$$u_{n+1} + u_n u'_{n+1} = K * u_n \quad (2.5)$$

and $u_n(-\infty) = u_c$. This iteration has the same properties on $(-\infty, 0)$ and $(0, \infty)$, by odd extension and the evenness of K . In the inductive step we show that the statement $\exists$ continuous u_n such that $u_n > 0$, $u_n(-\infty) = u_c$, $u'_n \leq 0$, $u_n - u_{n-1} \leq 0$, $u_0 \geq u_n \geq s$ implies the statement $\exists$ continuous u_{n+1} such that $u_{n+1} > 0$, $u_{n+1}(-\infty) = u_c$, $u'_{n+1} \leq 0$, $u_{n+1} - u_n \leq 0$, $u_0 \geq u_{n+1} \geq s$.

The solution to the linear equation (2.5) with the initial condition $u_{n+1}(0) = u_{n+1}^0 > 0$ and $x < 0$ is given by the formula

$$u_{n+1}(x) = e^{\int_x^0 \frac{dy}{u_n(y)}} \left[u_{n+1}^0 - \int_x^0 e^{-\int_y^0 \frac{dr}{u_n(r)}} \frac{K * u_n(y)}{u_n(y)} dy \right]. \quad (2.6)$$

However, u_{n+1} in (2.6) is such that $u_{n+1}(-\infty) = u_c$ only if we choose

$$u_{n+1}^0 = \int_{-\infty}^0 e^{-\int_y^0 \frac{dr}{u_n(r)}} \frac{K * u_n(y)}{u_n(y)} dy, \quad (2.7)$$

which follows from l'Hôpital's rule:

$$\lim_{x \rightarrow -\infty} \frac{u_{n+1}^0 - \int_x^0 e^{-\int_y^0 \frac{dr}{u_n(r)}} \frac{K * u_n(y)}{u_n(y)} dy}{e^{\int_x^0 \frac{dy}{u_n(y)}}} = \lim_{x \rightarrow -\infty} K * u_n(x) = u_c.$$

Note that the integral in (2.7) is convergent, since $u_n \leq u_c$ implies $e^{-\int_y^0 \frac{dr}{u_n(r)}} \leq e^{\frac{y}{u_c}}$. Let $v_{n+1} = u_{n+1} - u_n$, $v_n = u_n - u_{n-1}$. From (2.5) we get

$$v_{n+1} + u_n u'_{n+1} - u_{n-1} u'_n = K * v_n.$$

After adding and subtracting $u_n u'_{n+1}$ on the left side we get

$$v_{n+1} + u_n v'_{n+1} + u'_n v_n = K * v_n. \quad (2.8)$$

Since v_n is odd, $K * v_n$ can be represented on $(-\infty, 0)$ as

$$K * v_n(x) = \int_{-\infty}^0 [K(x-y) - K(x+y)] v_n(y) dy. \quad (2.9)$$

Notice that $K(x-y) - K(x+y) \geq 0$ for $x \leq 0$ and $y \leq 0$ because K is even and nondecreasing for $x < 0$, and thus $K * v_n \leq 0$. If $v_{n+1} > 0$ somewhere, then since $v_{n+1}(-\infty) = 0$, $\exists x_0$ such that $v_{n+1}(x_0) > 0$ and $v'_{n+1}(x_0) > 0$. Since $v_n \leq 0$, $u'_n \leq 0$, the left side of (2.8) is positive, a contradiction. Now differentiate (2.5):

$$u'_{n+1} + u'_n u'_{n+1} + u_n u''_{n+1} = K' * u_n = -K(x)[u_n(0-) - u_n(0+)] + K * u'_n \leq 0. \quad (2.10)$$

If $u'_{n+1} > 0$ somewhere, then $\exists x_0$ such that $u'_{n+1}(x_0) > 0$ and $u''_{n+1}(x_0) > 0$. The left side of (2.10) at x_0 is then positive, since from (2.5), $u'_n(x_0) > -\frac{u_n(x_0)}{u_{n-1}(x_0)} \geq -1$, a contradiction. Using $u'_{n+1} \leq 0$ we get $u_{n+1} \geq u_{n+1} + u_n u'_{n+1} = K * u_n > 0$ for $x < 0$.

Assuming $u_n \leq u_0$ we get

$$u_{n+1} + u_0 u'_{n+1} \leq u_{n+1} + u_n u'_{n+1} = K * u_n \leq K * u_0 = u_1 + u_0 u'_1,$$

and thus

$$u_{n+1} - u_1 + u_0(u'_{n+1} - u'_1) \leq 0,$$

and, since $u_{n+1}(-\infty) = u_0(-\infty)$, $u_{n+1} \leq u_0$ by an argument similar to the above. Assuming $u_n \geq s$, in the same way we get

$$u_{n+1} - u_1 + s(u'_{n+1} - u'_1) \geq 0$$

and $u_{n+1} \geq s$.

Thus $\{u_n\}$ is a sequence of monotone functions such that $u_{n+1}(x) \leq u_n(x)$ for all $n \geq 0$ and $x < 0$. Let $u \equiv \lim_{n \rightarrow \infty} u_n$. Since $u_0 \geq u_n \geq s$ for all $n \geq 0$, $u(-\infty) = u_c$. In (2.5) add and subtract $u_{n+1}u'_{n+1}$, multiply it by $\phi \in C^\infty$, with ϕ compactly supported in $(-\infty, 0)$, and integrate over $(-\infty, 0)$:

$$\int_{-\infty}^0 \left[-\frac{1}{2}u_{n+1}^2\phi' + u_{n+1}\phi - u'_{n+1}(u_{n+1} - u_n)\phi - (K * u_n)\phi \right] = 0. \quad (2.11)$$

As above, from (2.5) $u'_n > -1$, and thus $|u'_n|$ is uniformly bounded in n and we can use the Lebesgue Dominated Convergence Theorem in (2.11) to pass to the limit as $n \rightarrow \infty$ to get

$$\int_{-\infty}^0 \left[-\frac{1}{2}u^2\phi' + (u - K * u)\phi \right] = 0.$$

Since $\frac{1}{2}u^2 \in W^{1,\infty}(-\infty, 0)$, $\frac{1}{2}u^2$ is differentiable a.e. and $(\frac{1}{2}u^2)' = K * u - u$ a.e. on $(-\infty, 0)$. $\frac{1}{2}u^2 \in W^{1,\infty}(-\infty, 0)$ also implies that $\frac{1}{2}u^2$ and therefore u is Lipschitz continuous, and thus $u \in C^1(-\infty, 0)$ and u satisfies (2.1) on $(-\infty, 0)$. Since u is odd, it also satisfies (2.1) on $R \setminus 0$ and (1.4). To show $u'(x) < 0$ for $x \neq 0$, suppose there is a point x_0 such that $u'(x_0) = 0$. Then x_0 belongs to an interval I in which $u' = 0$. Let $\bar{x}_0$ be its right end point ($x_0 = \bar{x}_0$ if $I = \{x_0\}$). Differentiating (2.1) we get

$$u' + u'^2 + uu'' = -K(x)[u(0-) - u(0+)] + K * u'.$$

Since at $\bar{x}_0$, $u'' = 0$ and the right side is negative, we get a contradiction. If $u'' \geq 0$ somewhere on $(-\infty, 0)$, then there is a point x_0 such that $u''(x_0) = 0$ and $u'''(x_0) \geq 0$. Differentiating (2.1) twice we get

$$u'' + 3u'u'' + uu''' = -K'(x)[u(0-) - u(0+)] + K' * u'.$$

At x_0 the left side is nonnegative and the right side negative, a contradiction. If u is continuous, then since $u' > -1$ and u' is decreasing on $(-\infty, 0)$, u is also C^1 .

To find a condition for which u is discontinuous, we argue by contradiction. Assume that u is continuous at 0. Integrating (2.1) over $(-\infty, 0)$ and using Fubini's Theorem and then the LDC Theorem we get

$$\begin{aligned} -\frac{1}{2}u_c^2 &= \int_{-\infty}^0 uu' = \int_{-\infty}^0 (K * u - u) = \int_{-\infty}^0 \int_R K(y) \int_0^1 yu'(x + yt) dt dy dx \\ &= \int_R yK(y) \int_0^1 \lim_{m \rightarrow -\infty} [u(yt) - u(-m + yt)] dt dy = \int_R yK(y) \int_0^1 u(yt) dt dy. \end{aligned}$$

The right side can be estimated to yield

$$\frac{1}{2}u_c^2 \leq u_c \int_R |y|K(y) dy.$$

Thus if $u_c > 2 \int_R |y|K(y) dy$, we obtain a contradiction. This gives the criterion for a solution to be discontinuous as $u_- - u_+ > 4 \int_R |y|K(y) dy$, since $u_c = \frac{1}{2}(u_- - u_+)$. For $K(y) = \frac{k}{2}e^{-k|y|}$, we get $u_- - u_+ > \frac{4}{k}$. $\square$

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